

UnJin Lee, PhD

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Positions

- 2024- Postdoctoral Fellow, Zhao Lab, The Rockefeller University
 National Science Foundation Postdoctoral Research Fellowship in Biology
- 2022-2024 Postdoctoral Associate, Zhao Lab, The Rockefeller University

Education

PhD, MS University of Chicago

Committee on Genetics, Genomics, and Systems Biology, 12/2021

On the Mechanisms of Genomic and Phenotypic Evolution

Advisor: Manyuan Long

BA, University of Chicago

Physics, 2013

Research Experience

Publications

- C. Thompson, S. Mozeika, E. Paredes, **U. Lee**[†], Korean Natural Farming practices are dominated by a limited number of microbes and decrease fungal diversity. (to be submitted at Journal of Soil Science and Plant Nutrition) [preprint](#)
- U. Lee**, C. Li, C. B. Langer, N. Svetec, L. Zhao, Comparative single cell analysis of transcriptional bursting reveals the role of genome organization on de novo transcript origination. *PNAS*, **122**(18):e2425618122 (2025) [pubmed](#)
- U. Lee**[†], D. Arsala^{*}, S. Xia^{*}, C. Li, M. Ali, N. Svetec, C.B. Langer, D. Sobreira, I. Eres, D. Sosa, J. Chen, L. Zhang, P. Reilly, A. Guzzetta, J.J. Emerson, P. Andolfatto, Q. Zhou, L. Zhao, M. Long. The 3-Dimensional Genome Drives the Evolution of Asymmetric Gene Duplicates via Enhancer Capture-Divergence. *Science Advances*, **10**, eadn6625 (2024) [pubmed](#)
- U. Lee**, S. Mozeika, L. Zhao, A Synergistic, Cultivator Model of De Novo Gene Origination. *Genome Biology and Evolution*. evae103 (2024). [pubmed](#)
- U. Lee**[†], E. N. Mortola, M. Long, E. Kim. Evolution and maintenance of phenotypic plasticity. *BioSystems*. **222**, 104791 (2022). [pubmed](#)
- S. Xia, N. W. VanKuren, C. Chen, L. Zhang, C. Kemkemer, Y. Shao, H. Jia, **U. Lee**, A. S. Advani, A. Gschwend, M. D. Vibranovski, S. Chen, Y. E. Zhang, M. Long, Genomic analyses of new genes and their phenotypic effects reveal rapid evolution of essential functions in *Drosophila* development. *PLOS Genetics*. **17**, e1009654 (2021). [pubmed](#)
- J. Zu, Y. Gu, Y. Li, C. Li, W. Zhang, Y. Zhang, **U. Lee**, L. Zhang, M. Long, Topological evolution of coexpression networks by new gene integration maintains the hierarchical and modular structures in human ancestors. *Science China Life Sciences*. **62** (2019). [pubmed](#)
- E. Leypunskiy, J. Lin, H. Yoo, **U. Lee**, A. R. Dinner, M. J. Rust, The cyanobacterial circadian clock follows midday in vivo and in vitro. *eLife*. **6**, e23539 (2017). [pubmed](#)
- E. Kim, **U. Lee**, J. Heseltine, R. Hollerbach, Geometric structure and geodesic in a solvable model of nonequilibrium process. *Phys. Rev. E*. **93**, 062127 (2016). [pubmed](#)

U. Lee[†], J. J. Skinner, J. Reinitz, M. R. Rosner, E.-J. Kim, Noise-driven phenotypic heterogeneity with finite correlation time in clonal populations. *PLOS ONE*. **10**, e0132397 (2015). [pubmed](#)

U. Lee*, C. Frankenberger*, J. Yun, E. Bevilacqua, C. Caldas, S.-F. Chin, O. M. Rueda, J. Reinitz, M. R. Rosner, A prognostic gene signature for metastasis-free survival of triple negative breast cancer patients. *PLOS ONE*. **8**, e82125 (2013). [pubmed](#)

*contributed equally

[†]corresponding author

Presentations

- | | |
|------|---|
| 2025 | <u>66th Annual Drosophila Research Conference</u>
Single-cell research in Drosophila: technologies and resources workshop
San Diego, CA
<i>scHSQ allows for the alignment of evolutionarily diverged tissue types (invited workshop talk)</i> |
| 2024 | <u>NY Area Population Genetics Meeting</u>
New York City, NY
<i>Comparative scRNA-Seq analysis of Drosophila spermatogenesis reveals critical role of genome organization in de novo transcript origination (selected talk)</i> |
| 2023 | <u>SMBE Satellite Meeting on De Novo Genes</u>
College Station, TX
<i>Origination of a Testis-Specific lncRNA (poster, best presentation award)</i> |
| 2022 | <u>NY Area Population Genetics Meeting</u>
New York City, NY
<i>Evolution and Maintenance of Phenotypic Plasticity (poster)</i>
<i>The Role of the 3D Genome in New Gene Evolution (poster)</i> |
| 2019 | <u>Midwest Population Genetics Meeting</u>
Chicago, IL
<i>Evolutionary Consequences of Phenotypic Diversification (selected talk)</i> |
| 2019 | <u>Gordon Conference on Ecological and Evolutionary Genomics</u>
Manchester, NH
<i>Regulatory Evol. of a New Essential Gene is Driven by an Old Enhancer (poster)</i> |
| 2016 | <u>Aspen Center for Physics on Evolution, Populations, and Physics</u>
Aspen, CO
<i>Selection Over a Model of Noise-Driven Phenotypic Heterogeneity (poster)</i> |
| 2013 | <u>American Association of Cancer Researchers Annual Conference</u>
Washington, DC
<i>A Prognostic Gene Sig. for Metastasis-Free Survival of TNBC Patients (poster)</i> |

Awards and Honors

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|-----------|---|
| 2024-2027 | NSF Postdoctoral Research Fellowship in Biology Recipient |
| 2023 | Best Presentation Award, SMBE Satellite Meeting on De Novo Genes |
| 2023 | SMBE Satellite Meeting on De Novo Genes Travel Award |
| 2017-2021 | Ridgeway Endowment Support |
| 2017 | Hinds Evolutionary Biology Graduate Student Research Award |
| 2016 | NSF Graduate Research Fellowship Program Honorable Mention |
| 2015-2017 | NIH T32 Gene Regulation Training Grant Recipient |
| 2011 | NSF Res. Experiences for Undergraduates, Chicago Center for Systems Biology |
| 2008-2013 | Odyssey Scholarship Recipient, University of Chicago |

Service Activities

Outreach

Summ. 2023	RockEDU SSRP Mentor
Spring 2023	Genspace FIT Scholar Program Mentor
2017-2019	EE Just Portrait Commission
2016-2018	Co-Host Groks Science Radio Hour WHPK 88.5FM
2015-2016	Genetics Science Connections at Museum of Science and Industry

Professional

2023-	Rockefeller University Sustainability Committee Laboratory Representative
2023-2024	Rockefeller University Post-Doc Association Board Member
2022-	Rockefeller Inclusive Science Initiative Executive Board Member (Mental Health and Accessibility Coordinator 2023-)
2017-2018	BSD Equipment Library Committee (Founder)
2017-2019	BSD Travel Award Committee (Member, Chair 2018-2019)
2014-2019	Dean's Council Representative
2014-2016	Program Representative (GGSB)

Teaching

Instruction

Fall 2023	Workshop Instructor, Genspace (<i>"Introduction to Bioinformatics"</i>)
Spring 2023	Workshop Instructor, RockEDU SSRP (<i>"Introduction to Molecular Evolution"; "Detecting Signatures of Selection"; "Genetic Sources of Phenotypic Diversity"</i>)
Spring 2023	Workshop Instructor, Genspace (<i>"Introduction to Bioinformatics"</i>)
Winter 2018	Grader, University of Chicago (<i>"Fundamentals of Genetics"</i>)
Spring 2017	Teaching Assistant, University of Chicago (<i>"Classics of Evolutionary Genetics"</i>)
Winter 2017	Grader, University of Chicago (<i>"Fundamentals of Genetics"</i>)
Fall 2016	Teaching Assistant, University of Chicago (<i>"Developmental Biology"</i>)

Trainees

2024-	Candance Thompson (<i>"Microbial Community Propagation Through Korean Natural Farming Practices Decreases Fungal Diversity and Increases Microbial Diversity"</i>)
Summ. 2023	Yifei Zhou (<i>"Ablation & Genesis: Comparative Genetic Analysis of Behavioral Preference in an Invasive Pest Species"</i>)
Summ. 2023	Finn Charest (<i>"Randomness is the Spice of Life: Examining Stochastic Fluctuations in Lotka-Volterra Models"</i>)
Summ. 2023	Bharat Elango (<i>"Machine Learning-Based Analysis of Gene Expression in D. melanogaster"</i>)
Spring 2023	Kiera Derrig (<i>"Curating My Arsenal of Colorants with Bacteria"</i>)
Summ. 2019	Gustavo Rangel T. de Almeida (<i>"Numerical Optimization Methods in Python"</i>)
2017-2019	Jack Mortola (<i>"Regulatory Evolution of a New Essential Gene is Driven by an Old Enhancer"</i>)

Professional Skills Development

Summ. 2025	Academic Job Search Boot Camp <i>Participant</i>
Summ. 2024	Tri-Institutional Transitioning to Research Independence Series <i>Participant</i>
Summ. 2024	Managing Science and Scientists Workshop <i>Invited Alumni Panelist</i>

Summ. 2023 Managing Science and Scientists Workshop
*Three day management workshop taught by Harvard Business School and Haas School
of Business faculty*
Workshop Participant (Inaugural)

Scientific Software

[sigsquared](#) (Bioconductor/R package)

[scHSQ](#) (R package)

Intellectual Property

[Prognostic and predictive breast cancer signature](#) (M. Rosner, M. Sun, **U. Lee**, USPTO
14/952,265)

Peer Review

PLoS Computational Biology, Cancer Cell International (Springer), National Science
Foundation (*ad hoc* reviewer), PNAS